IGRIS Teacher Policy — Training Environment Summary

Task: TWIST2-IGRIS-Priv-Teacher-Robust-Direct-v0 | Run: v3_motion_curriculum | 2026-03-24

| 1. Training Overview

Item	Value
Task	TWIST2-IGRIS-Priv-Teacher-Robust-Direct-v0
Algorithm	PPO (OnPolicyRunner, RSL-RL)
Num Envs	4,096
Max Iterations	100,000 (v3 resumed from iter 7,600)
Sim / Step dt	0.002 s / 0.02 s (decimation=10)
Episode Length	500 steps = 10 s
Run name (v3)	v3_motion_curriculum — motion_curriculum=True, level_curriculum=False

| 2. PPO Hyperparameters

Parameter	Value	Note
num_steps_per_env	64	rollout buffer length per env
num_learning_epochs	5	
num_mini_batches	4	~67K samples per mini-batch
learning_rate	1e-4	adaptive (desired_kl=0.008)
clip_param	0.2	
entropy_coef	*0.01*	** modified: 0.0 -> 0.01 (prevent entropy collapse)*
value_loss_coef	1.0	
gamma / lambda	0.99 / 0.95	
max_grad_norm	1.0	

| 3. Network Architecture

Module	Hidden Dims	Activation	Note
Actor	*[512, 512, 256, 128]*	*SiLU*	** modified: ELU -> SiLU*
Critic	[512, 512, 256, 128]	SiLU	obs_normalization=True
Obs Norm	EmpiricalNormalization	—	running mean/std
Distribution	GaussianDistribution	init_std=1.0	

| 4. Observation Space (total 2,764 dim)

Group	Dim	Contents	Noise
(1) Motion Future Obs	1,580	20 future timesteps x 79 dim ref_root_pos(3) / pos_delta(3) / rpy(3) / vel_local(3) ang_vel_local(3) / pos_delta_local(3) / rot_delta_local(3) dof_pos(31) / key_body_local(27)	Yes

Group	Dim	Contents	Noise
(2) Proprio Obs	98	ang_vel_b x0.25(3) / roll-pitch(2) joint_pos_offset x1.0(31) / joint_vel x0.05(31) actions(31) [raw_actions if action_delay active]	Yes
(3) Privileged Info	106	root_lin_vel_b(3) / root_pos_w(3) / root_quat_w(4) key_body_local_cur(27) / foot_contacts(2) mass_feature(4) / friction_feature(1) / motor_offset(62)	No
(4) Proprio History	*980*	** NEW: 10 timesteps x 98 dim (sliding window of proprio)* *reset to zero on episode start*	*Yes*

| 5. Reward Structure

Tracking rewards (positive)

Term	Scale	Formula
track_root_pos	1.0	exp(-4 * ||pos_err||^2) * dt
track_root_rot	0.75	exp(-2 * rot_err^2) * dt
track_root_vel	0.5	exp(-2 * ||vel_err||^2) * dt
track_root_ang_vel	0.5	exp(-0.5 * ||ang_vel_err||^2) * dt
track_dof_pos	1.0	exp(-0.25 * ||dof_err||^2) * dt
track_dof_vel	0.35	exp(-0.05 * ||dof_vel_err||^2) * dt
track_key_body	1.5	* modified: 0.75->1.5 exp(-2 * key_body_local_err) * dt
track_key_body_global	*1.5*	** modified: 0.75->1.5 exp(-2 * key_body_global_err) * dt*
alive	*0.25*	*constant * dt*

Penalty terms (negative)

Term	Scale	Formula
action_rate	-0.01	||a - a_prev||^2 * dt
torque_penalty	-1e-6	cmd override (cfg default: -1e-5) ||tau||^2 * dt
feet_slip	-0.1	slip_velocity * dt
feet_contact_forces	-5e-4	excess_contact_force * dt
feet_air_time	+5.0	(air_time - 0.5s).clamp(max=0) * dt [always <= 0]
feet_stumble	-1.25	* dt
lin_vel_z	-1.0	vz^2 * dt
orientation	-1.0	projected_gravity_xy^2 * dt
ang_vel_xy	-0.01	||omega_xy||^2 * dt
dof_vel	-1e-4	||joint_vel||^2 * dt
ankle_dof_vel	-2e-4	||ankle_vel||^2 * dt
joint_acc	-5e-8	||joint_acc||^2 * dt
dof_pos_limits	-5.0	limit_violation * dt
dof_torque_limits	-1.0	torque_limit_violation * dt
collision	-10.0	penalized_contact * dt
terminated	-2.0	terminated * dt

| 6. Domain Randomization

Item	Setting	Note
Mass scale	x(0.9 ~ 1.1)	resampled on reset

Item	Setting	Note
Joint friction	x(0.75 ~ 1.25)	resampled on reset
Stiffness / Damping	x(0.85 ~ 1.15)	resampled on reset
Root CoM offset	+/- 0.05 m	resampled on reset
Gravity XY	+/- 0.1 m/s^2	resampled every 4 s
Robot push	disabled	--enable_teacher_robot_push=false
End-effector push	disabled	--enable_teacher_end_effector_push=false
Action delay	0~1 step	activated after obs_noise ramp
Obs noise	*0 -> max (ramp)*	** modified: ramp over 10,000x24 steps (~iter 11,355 = 100%)*

Obs noise max values (at noise_progress=1.0):

Obs term	Max noise
root_pos / key_body	0.02
orientation / IMU	0.05
root_vel / root_ang_vel / dof_vel / ang_vel	0.1
dof_pos	0.01

| 7. Curriculum

Item	Setting	Note
motion_curriculum	*True * v3 new*	*per-motion adaptive difficulty*
motion_curriculum_gamma	0.01	difficulty update rate
difficulty update rule	completion <= 50% -> x(1+gamma) completion >= 95% -> x(1-gamma) completion >= 99% -> x(1-20*gamma)	
difficulty range	1.0 ~ 10.0	initialized to 10.0 for all motions
enable_motion_level_curriculum	*False * v3 changed*	*level gating disabled -> levels 1-5 all open*
use_error_aware_sampling	True	key body error weighted sampling

| 8. Termination Conditions

Condition	Threshold
Root height	< 0.65 m
Body tilt	projected_gravity_xy^2 > 0.8
Torso contact	contact on body_link
NaN / Inf state	joint pos/vel/acc, contact forces, etc.
Timeout	500 steps (10 s)

| 9. Changes from v1 (original)

Item	v1 (original)	v3 (current)	Effect
entropy_coef	0.0	0.01	prevent entropy collapse
activation	ELU	SiLU	smoother gradient
num_steps_per_env	32	64	better credit assignment
obs_noise_ramp_steps	3,000	10,000	gradual noise introduction
proprio_history_len	0	10	noise/delay compensation
track_key_body scale	0.75	1.5	stronger keypoint tracking

Item	v1 (original)	v3 (current)	Effect
motion_curriculum	False	True	focus on hard motions
level_curriculum	True	False	remove cumulative counter issue

Generated from twist2_lab codebase — 2026-03-24